# CAP 6938 - AI Knowledge Systems (Fall 2026): Project Proposal - STL Generating Agent

Michael Laager
University of Central Florida
Department of Computer Science
mi740165@ucf.edu

## 1. Overview

Throughout multiple domains, the ability to generate a 3D representation of an object serves as a foundational step for performing computer aided design (CAD) tasks. The industry-standard representation for a 3D model is an STL file. Thus, speeding up the creation process, especially for a user who is not adept at using CAD software, can increase productivity.

The agent being proposed should be able to take a user query and produce an STL file as a result that matches the query. Some examples of possible queries involve:

- "Make me a ruler with my name engraved on it."
- "Create a tree that has a cat on it".
- "Make a die that has 2 snakes and 4 ladders instead of numbers."

The agent should be able to ask any discretionary questions, if applicable, ensuring the best possible result for the user. If the agent cannot directly generate an STL, it should try to find one on the internet for integration and use.

## 2. Technologies

The agent will need an LLM front-end to serve as the chat interface for the user. LLMs generally suffer from directly creating 3D representations, so the LLM will instead generate code as an intermediary step. For this project, Python integration with the CADQuery library will serve as the step from LLM $\rightarrow$ STL. As code generation may not be practical for complex objects, the LLM should also decide the possibility of querying existing STLs on the internet as an alternative.

The use of a public API will make the fetching of online STL files easier. It appears that Thingiverse has API access [1] for developers, so this is likely the API endpoint that will be used for online access. Though it is noted that the public API access is not necessarily required for STL generation, only the enhancement of potential results.

## 3. Overall Goals

The agent should be able to accomplish the following:

1. There should be a chat session with the user, allowing for the collection and refinement of relevant details for the STL creation.
2. The agent then orchestrates the creation and/or downloading of STL files from the internet.
3. The agent combines these in a meaningful way to generate a final STL.

## 4. Benchmark Strategy

To quantitatively assert the quality of STL generation, an LLM-as-judge paradigm can be introduced for benchmarking. In this system, a MLLM model will assess an image of the generated STL from multiple angles. If the MLLM can attest that the generated image is in fact similar to the initial query, then that STL is considered to be passing. Otherwise, it will not be. The similarity metric generated by the judge can be used as a metric for a pass-fail grade.

## 5. Estimated Timeline

| Wk | Dates | Focus / Milestone |
|---|---|---|
| 1+2 | 9/21–10/4 | CadQuery & LLM setup, crude prompt→STL integration |
| 3+4 | 10/5–10/18 | Few-shot example library + RAG retrieval; measure success-rate gain |
| 5+6 | 10/19–11/1 | LLM-as-judge via mesh rendering + MLLM scoring |
| 7+8 | 11/2–11/15 | Thingiverse API integration and integration testing |
| 9+10 | 11/16–End | Report generation and final improvements or enhancements |

## References

[1] MakerBot Industries. Thingiverse API Documentation (Swagger). https://www.thingiverse.com/developers/swagger. Accessed: Sep. 18, 2026. 1